Topic 8:Energy Resource



Global Energy Statistics 2016

Total energy Production	China (2640 Mtoe)
Total energy Consumption	China (3101 Mtoe)
Crude oil Production	Saudi Arabia (565 Mt)
Oil products Production	USA (868 Mt)
Oil products Consumption	USA (751Mt)
Natural Gas Production	USA (769 bcm)
Natural Gas Consumption	USA (783 bcm)
Coal, lignite Production	China (3538 Mt)
Coal, lignite Consumption	China (3732 Mt)
Electricity Production	China (5682 TWh)
Electricity Consumption	China (4921 TWh)



Types of energy Resources

- NON- RENEWABLE SOURCE
- RENEWABLE SOURCE

Types of fossil fuel

Coal, oil (petroleum), and natural gas are considered nonrenewable because they can not be replenished in a short period of time. These are called fossil fuels.







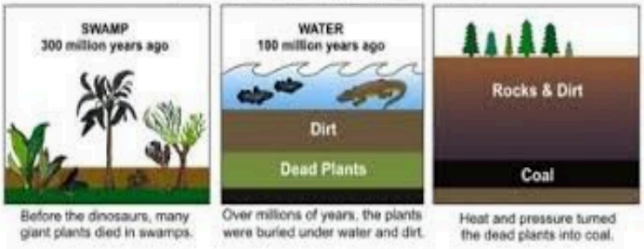


Coal Facts

- ✓ Most abundant fossil fuel produced in the United States.
- ✓ A nonrenewable energy source because it takes millions of years to create.
- ✓ Energy in coal comes from the energy stored by plants that lived hundreds of millions of years ago.

Formation of coal

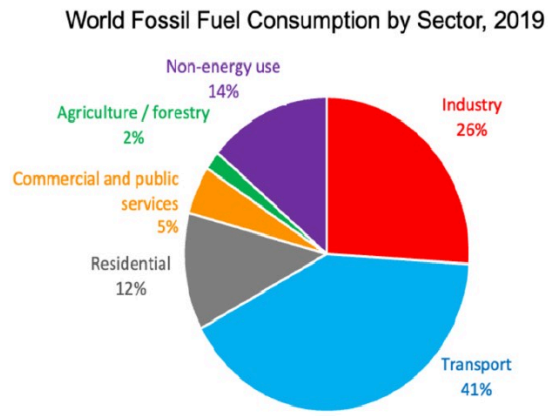
- Formed when peat is altered physically and chemically.
- Formed from the remains of vegetation that grew as long as 400 million years ago.
- Process is called "coalification."



Formation of natural gas

- Methane, a gas or compound that has one carbon atom and four hydrogen atoms.

Fossil Fuel Consumption sector



Uses of Coal

- Industries.
- Domestic Use.
- Gasification and Liquefaction.
- Almost 93% of the coal used in the United States is used for generating electricity.

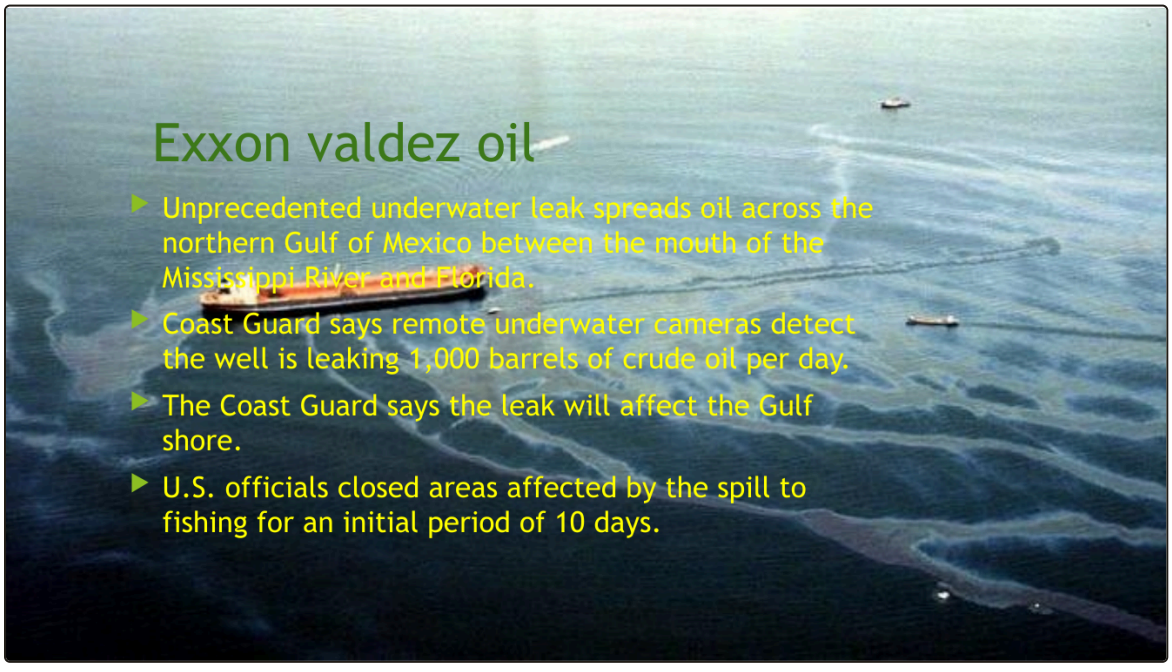
Uses Of Natural Gas

- Raw material to produce chemicals, fertilizer, and hydrogen.
- To heat the building and to dry clothes.
- To operate refrigeration and cooling equipment.
- For transportation services.



Disadvantages of fossil fuels

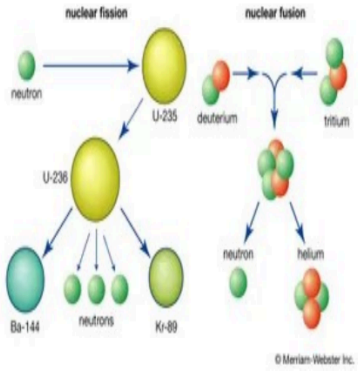
- Contribute to climate change.
- Non-renewable.
- Unsustainable.
- Hard to extract.
- Accident-prone.



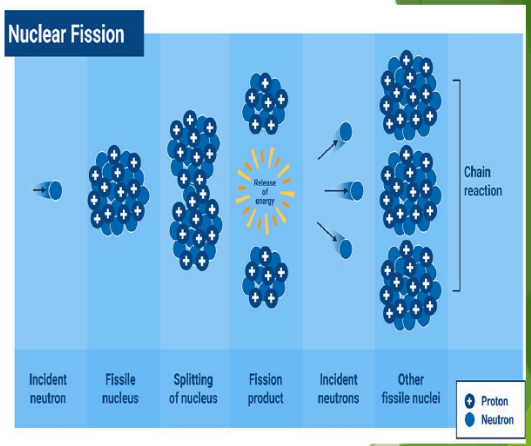
Exxon valdez oil

- Unprecedented underwater leak spreads oil across the northern Gulf of Mexico between the mouth of the Mississippi River and Florida.
- Coast Guard says remote underwater cameras detect the well is leaking 1,000 barrels of crude oil per day.
- The Coast Guard says the leak will affect the Gulf shore.
- U.S. officials closed areas affected by the spill to fishing for an initial period of 10 days.

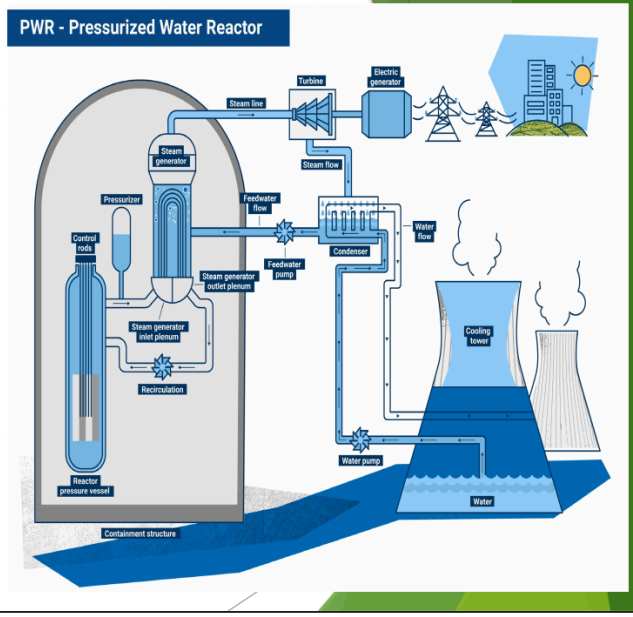
FISSION AND FUSION OF NUCLEAR ATOM



- In nuclear fission, for instance, when hit by a neutron, the nucleus of an atom of uranium-235 splits into two smaller nuclei, for example, a barium nucleus and a krypton nucleus, and two or three neutrons.
- These extra neutrons will hit other surrounding uranium-235 atoms, which will also split and generate additional neutrons in a multiplying effect, thus generating a chain reaction in a fraction of a second. Each time the reaction occurs, there is a release of energy in the form of heat and radiation. The heat can be converted into electricity in a nuclear power plant, similarly to how heat from fossil fuels such as coal, gas, and oil is used to generate electricity.
- Nuclear fusion is the process by which nuclear reactions between light elements form heavier elements. In cases where the interacting nuclei belong to elements with low atomic numbers (e.g., hydrogen [atomic number 1] or its isotopes deuterium and tritium), substantial amounts of energy are released.
- The vast energy potential of nuclear fusion was first exploited in thermonuclear weapons, or hydrogen bombs, which were developed in the decade immediately following World War II.



- The process may take place spontaneously in some cases or may be induced by the excitation of the nucleus with a variety of particles (e.g., neutrons, protons, deuterons, or alpha particles) or with electromagnetic radiation in the form of gamma rays.
- In the fission process, a large quantity of energy is released, radioactive products are formed, and several neutrons are emitted. These neutrons can induce fission in a nearby nucleus of fissionable material and release more neutrons that can repeat the sequence, causing a chain reaction in which many nuclei undergo fission, and an enormous amount of energy is released.
- If controlled in a nuclear reactor, such a chain reaction can provide power for society's benefit. If uncontrolled, as in the case of the so-called atomic bomb, it can lead to an explosion of awesome destructive force.



An example of a nuclear Reactor

- Inside nuclear power plants, nuclear reactors and their equipment contain and control the chain reactions, most fueled by uranium-235, to produce heat through fission. The heat warms the reactor's cooling agent, typically water, to produce steam. The steam is then channeled to spin turbines, activating an electric generator to create low-carbon electricity.

References

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